

Evaluation of Electrocardiogram Signals Classification Using CNN, SVM, and LSTM Algorithm: A review

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Abstract—The non-stationary signals of Electrocardiogram (ECG) are widely utilized to assess heartbeat rate and tune the major goal of this study is to give an overview of ECG classification Machine learning and neural network methods are employed. Furthermore, the major stage in ECG classification is feature extraction, which is used to identify a group of important characteristics that may achieve the highest level of accuracy. The optimization approach is used in conjunction with classifiers to get the optimal value for Its discriminant purpose was best served by using classifying parameters that best fit the discriminant purpose. Finally, this study evaluates the signal classification for ECG heartbeat using a Convolution Neural Network (CNN), Support Vector Machine (SVM), and Long Short Term Memory (LSTM), compare between them and present that the best method is LSTM for these cases based on the dataset. The author is certain that this study would be beneficial to researchers, scientists, and Engineers who operate in this field to discover relevant references.

Keywords — *Electrocardiogram, ECG, MIT-BIH Data Set, Classification, Deep Learning.*

I. INTRODUCTION:

According to subsequent WHO (World Health Organization) reports, Non-disease deaths are mostly caused by cardiovascular disease. Yearly, it claims the lives of an estimated 17.9 million people, accounting for 44% of all non-communicable disease fatalities worldwide [1]. A large percentage of cardiac captures were linked to cardiac arrhythmias, that are produced by abnormal electrical motivation organization and propagation across the myocardial tissue [2]. As a result, detecting heart arrhythmia early and easily was crucial for saving lives. An atrial sine node (SA node), which is located in the right atrium of the heart, regulates its heart rhythm. Electrical activity in the heart creates electronic potential changes on the skin's surface, that may be detected using electrodes attached to the patient's body and visually recorded inside an ECG. The electrocardiogram (ECG) is a well-known method for non-invasively assessing a patient's heart activity. A typical normal ECG cycle is made up of numerous different parts (P wave, Q wave, R wave, S wave, T wave, and U wave). This is expressed by time-domain

characteristics like amplitude, duration, interval, and segment as depicted in Fig.1. Atrial and ventricular depolarization that occurs in tandem produces the character waves. Irregular ECG digital audio documents could be used to diagnose cardiology arrhythmias like premature ventricular contraction pulse (V), atrial premature beat (A), and right bundled branch frames because cardiac arrhythmias change the origin of the electrical voltage and the resulting biological diffusion within in the heart (R).

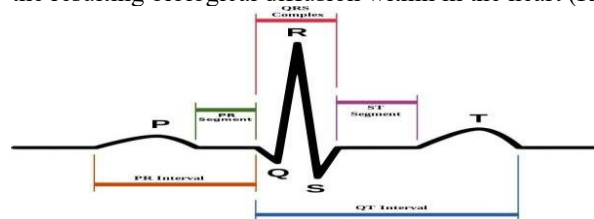


Fig .1 shows the waveform of an ECG signal. [45]

For the classed results, category annotation is used as the criteria. For segmenting ECG data, Annotation of the heartbeat locations by hand is used to place a sensor at the R-wave peak or major local extrema. Premature ventricular contraction (PVC), a fusion of ventricular and normal beats (F), atrial premature beat (A), right bundle branch block beat (R), and fusion of paced and normal beats (R) are all examples of normal beats. beat are six types which may be distinguished as (F).

An ECG waveform made up of five peaks may precisely characterize each heartbeat (features). P, T, Q, S, and R are the names of the summits. With the detection and evaluation of each peak, as a result of this, it is possible to identify the qualities of heart rhythm more effectively. and also their variations, distances, and some other mathematical factors. As a component of an automated ECG detection, Other characteristics of ECG signals, also as energy and entropy, were recovered as feature vectors using P, Q, R, S, and T wave patterns. the P, Q, R, S, and T

TABLE I. SHOWS THE CHARACTERISTICS OF THE ECG WAVEFORM.

Amount of amplitude (mV)	length of time (seconds)
P Wave - 0.25mV	PR interval - 0.12s to 0.20s
R Wave - 1.60mV	QT interval - 0.35s to 0.44s
Q Wave - 25% of R wave	ST interval - 0.05s to 0.15s T Wave 0.1 to
0.5mV	QRS interval - 0.09s

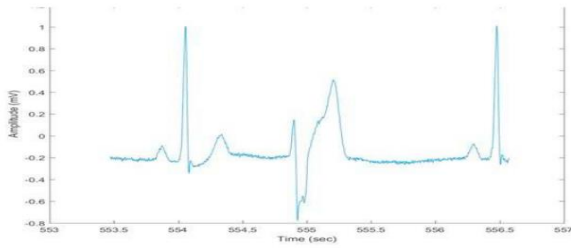


Fig.2 ECG Curve Plotted using MIT-BIH Data [48]

waves are a collection of data that describe the amplitudes and relative time intervals of such an ECG pulse signal. As illustrated in Table I. Fig.2 ECG Curve Plotted using MIT-BIH Data [48]

II. MIT-BIH ARRHYTHMIA DATA SET

The Harvard-MIT Division of Health Sciences and Technology, which is part of the Biomedical Engineering Center, is in charge of this database. It is made up of 48 half-hour data that were collected from 47 patients between 1975 and 1979. The records were digitized at 360 samples per second per channel, with an 11-bit resolution and a 10mV range. Each patient's file contains information such as the date of registration, age, gender, and drugs dispensed. The database may be found at <https://www.physionet.org/physiobank/database/mitdb/>. Arrhythmia dataset from the MIT-BIH dataset. This database is considered one of the most widely used databases for detecting and grouping arrhythmias, and it has been used extensively for algorithm evaluation. The recording as in the MIT-BIH electrophysiology dataset comes with an annotation file that serves as an Each pulse has its reference, which includes the category of heartbeat locations. An MIT-BIH Arrhythmia Dataset may be shown in Fig. 3.

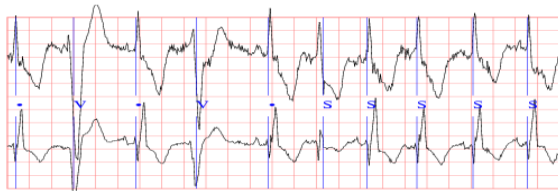


Fig.3 A Supraventricular Heart Rhythm Set of data from the MIT-BIH [47]

III. CLASSIFICATION

The ECG signal can provide a wealth of pathological information about a patient's cardiac functions [3]. Many methods for classifying heartbeats have been devised since it's crucial for detecting an arrhythmia [4]. Life-threatening vs quasi arrhythmias are the two types of arrhythmia. An ECG taken over a lengthy period categorization is required for quasi arrhythmia, which is a moment and inconvenient. As a result, automated ECG categorization of arrhythmias is one of the most interesting areas of research [5]. For the ECG classification problem, a

variety of classifiers have been utilized. The authors collect 143 papers and classified them into two classes before and after 2016. After the first filter 123 paper remains, the authors classified the remaining papers into three parts based on the methods applied for classification like (CNN, LSTM, and SVM). The authors selected 18 papers that applied the same dataset, and compare them. These classification methods may be grouped into numerous groups depending on the classifiers, such as CNN, SVM, and LSTM. Three well-known publishers, IEEE, Science Direct, and Springer, had access to all of the peer-reviewed publications. studying in a foreign country [5]. When there are so many signals in the ECG signal that may be retrieved, many categorization kinds approaches are investigated to categorize ECG data under variance features.

A. Classification Using Convolutional Neural Network (CNN)

A convolutional neural network's major parts are feature extraction and classification. The feature extraction component is in a duty of obtaining useful characteristics from The classification section is in charge of accurately classifying signals using the generated characteristics, while Signals are processed automatically. In a nutshell, these two parts work together to finish the major task of this study. A feature extraction technique includes layers such as convolution and also downsampled. Convolution (C-layer) is a sort of fuzzy filter that can improve signal qualities while lowering noise. Its feature vectors of the upper layer are convolution with the convolutional network of the current layer. Finally, the activation function returns the convolution computations' results. The error is computed using the following formula in comparison to the output vector that has been supplied as a goal. The training is completed if it is satisfied; in the meantime, the weights and thresholds are saved. Each weight is assumed to be stable, and the classifier is produced.

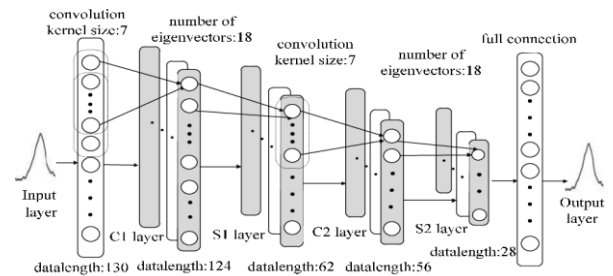


Fig. 4: Proposed deep neural networks structure [46]

As a result, the CNN model's structure must be adjusted. Figure 4 shows the CNN model for ECG classification that has been optimized. The structure consists of input, two fully connected layers, there are two down-sampled layers, as well as a full connection layer, and an output layer. Each neuron's input is coupled to the preceding layer's output, which is utilized to extract local characteristics. A C1 convolution

layer spreads 18 convolutions across a 7-sample range, with a 130-sample ECG signal segment as its input. In the sampling layer S1, the feature vectors from the C1 layer are pooled, and the feature vectors are also condensed into 62 sample points. A C2 layer likewise has 18 convolutional filters with such a length of 7 sample points and produces 324 feature vectors using 56 sampling points, while the C2 layer has 18 convolutional filters with a length of 7 sample sites and produces with 56 sample sites, 324 feature vectors were applied. When the S2 layer aggregated the feature vectors anew, the length was changed to 28 data points, they were then transmitted to the output layer, where the segmentation results were calculated.

The output of the convolution layer can be represented by “(1)” :

$$X_j^I = f \left(\sum_{i \in M_j} X_j^{I-1} * W_{ij}^I + b_j^I \right) \quad (1)$$

The spatial resolution between the hidden layer and hidden layer is decreasing and its formula is as below “(2)” :

$$X_j^I = f(B_j^I \text{down}(X_j^{I-1}) + b_j^I) \quad (2)$$

The error is calculated according to the following “(3)” compared with the given target output vector:

$$E = \frac{1}{2} \sum_{k=0}^{n-1} (d_k - y_k)^2 \leq \varepsilon \quad (3)$$

M.Zubair, J.Kim et al. (2016) suggested a model with two major components: feature extraction and classification. The algorithm uses raw ECG data to recall an appropriate feature representation, eliminating the requirement for hand-crafted features. The suggested classification method efficiently categorized ECG beats into five distinct groups using minimal and patient-specific training data. The suggested approach outperforms most region approaches for ECG signal classification in terms of classification accuracy and computing efficiency [6], and the findings show that the proposed approach outperforms most region methods in ECG signal categorization. W. Yin, X. Yang, et colleagues (2016) proposed a technique for categorizing arrhythmia of ECG recordings collected in the small motion condition using includes impulse radio super-duper radar data for additional information. A cascaded CNN is also used in this suggested system to accomplish an ECG study that is comprehensive records as well as radar data. In mild motion conditions, the experiments are carried out from the Caffe platform, with accuracy is 88.89%. It turns shown that in the mild motion condition, our suggested method maintains a stable accuracy of categorization for both regular and abnormal heartbeats [7]. S. L. Oh, N. A. Polytechnic et al. (2017) To achieve this, a 9- layer CNN DCNN was used to automatically detect 5 distinct types with ECG readings for heartbeats. ECG signals Were on collected from one database are both available. The set was artificially boosted to reduce large noise Heartbeat detection for sound and original Electrocardiography, the CNN was used to train the model to use supplemented data to achieved an Accuracy is 93.470 percent and 94.030 percent, respectively [8]. X.Zhai

and C.Tin. (2018) suggested a method based on the new CNN design The SEN of positive prediction rate for S pulses of a model both were increased by more than 12.2%,11.9%, correspondingly. Without any manual feature extraction or professional assistance, the system delivered an automated and reliable tool for detecting arrhythmia heartbeat [9]. J. Zhang, J. Tian, Y. Cao et al. (2019) used DCNN to develop a new pattern recognition algorithm for ECG data. The suggested technique learned features, and classification methods from of the time series, unlike prior approaches that employed from of the frequency response domains, learn features using hand-crafted characteristics. The ECG waveform signal first was converted to the time-frequency phase using the Long Transformation Function. Then, using ECG data of varying lengths, various scale-specific DCNN models were trained. Finally, an online decision fusion approach [10] was presented to fuse judgments from several transforms into one that is more accurate and reliable. J. Wang. (2020) developed a unique technique A DNN having 11 layers was used for automatic vehicle Atria fibrillation (AF)diagnosis. This network structure was merged with CNN, employing the modified Elman neural net. 10 cross-validation were performed to evaluate the model's generalization ability here on the MIT-BIH AF dataset. The Accuracy, SEN, and SPE of 97.4 percent, 97.9 percent, and 97.1 percent, respectively, demonstrated that the model had good classification performance [11]. Q. Yao, R. Wang, X. Fan et al. (2020) outlined show consideration based progressively over times CNN(ATI-CNN), DNN show might get both physically and worldly combination of data based on ECG signals utilizing joining CNN. The highlights were the adaptable length of the input, in real-time preparation, there is a split parameter total and a 90% reduction in the calculation. In comparison with VGGNET, a typical 16-layer CNN, ATI-CNN had an average classification performance of 81.2 percent, according to the analysis. ATI-CNN increased Accuracy by 7.7% in normal conditions as up 26.8% when detecting neurogenic arrhythmia [12]. Table II shows the comparison of the mentioned papers that used CNN.

B. Classification Using Support Vector Machine (SVM)

SVM is a learning method with a lot of advantages. It has to do with data analysis and pattern recognition. For classification, it employs a linear discriminate function [13] However, if a non-linear kernel is utilized, non-linear classification can be achieved [14]. SVM is a robust, easy-to-understand algorithm that performs well in real-time situations. It has a global solution when compared to other classifiers. SVM's performance is determined by the kernel function; more information on SVM can be found in [15]. SVM is a machine learning method with a lot of useful qualities. It is linked to data processing that is used to identify a design. SVM uses a direct separate work for classification; however, if a non-linear portion is used, non-linear categorization may be done as well [16]. SVM is fast in real-

time scenarios, and it's easy to learn. In contrast to other classifiers. A categorization assignment frequently necessitates knowledge about the topic to be categorized; as a result, the classifiers must be prepared to classify the information lately [17].

One of the most interesting aspects of the SVM classifier is that it automatically discovers bolster vectors for much better classification [18]. The execution of SVM is largely dependent on the affected bit work determination for each case [19]. F. A. Elhaj, N. Salim et al. (2016) looked at

TABLE II: CNN-BASED CLASSIFICATION OF HEARTBEAT ALGORITHMS

Author Name	DataBase	Purpose	Models	Accuracy
M.Zubair, J. Kim et al. (2016)	MIT-BIH arrhythmia data set	Proposed learn features from raw ECG	1D-CNN	Accuracy=92.7%
W. Yin, X. Yang (2016)	The data is based on the BMD101 ECG sensing element and the Bluetooth module.	ECG, radar signals are being monitored and classified.	Cascade CNN	Accuracy=88.89%
S. L. Oh, N. A. Polytechnic et al (2017)	MIT-BIH arrhythmia data set	Identified automatically five different categories of ECG	9-layers CNN	Accuracy With noise=94.030% Without
X. Zhai and C. Tin. (2018)	MIT-BIH arrhythmia data set	A model was applied on the portable device with long-term monitoring.	CNN	Accuracy>97%
J. Zhang, J. Tian, Y. Cao et al et al. (2019)	Synthetic real-world ECG data	The proposal uses the time-frequency domains to train features and classification methods.	DCNN	Accuracy=99%
J. Wang (2020)	MIT-BIH arrhythmia data set	A method for detecting AF automatically has been proposed.	CNN+MENN	Accuracy=97.4%
Q. Yao, R. Wang, X. Fan al. (2020)	China Physiologic al Challenge dataset	Differentiate between ECG signals of various lengths.	ATI-CNN (attention-based time-incremental)	Accuracy=81.20%

combining linear and non-linear variables to enhance ECG classification The study examines five types of arrhythmias beat categories, Non-ectopic beats (N), supraventricular ectopic beats (S), ventricular ectopic beats (V), fusion beats

(F), and unclassifiable and paced beats (U) are the classes used by the Association to Standardization in Clinical Instrumentation (U). Nonlinear characteristics like high-frequency statistics and cumulates, as well as non-linear feature reduction methods like ICA, are used with linear features, most notably the PCA of DWT coefficients, for boost categorization. Fold cross-validation tests the features' ability to discriminate multiple employing different types of data various categories, particularly the SVM and NN techniques. Using a combined SVM and Radial Basis Function RBF approach, with a high Accuracy (98.91%), Its N. S. V. F. and U arrhythmias types can be classified using this method [20]. Statistics aspects, according to V. R. Arjunan (2016), might be effective for classifying ECG signals. This de-noising operation has resulted in a signal that was pre-processed, just like the first. The signal is then processed to extract statistical properties such as mean, variance, standard deviation, and skewness. SVM was used to divide the ECG data into two groups: normal and pathological. According to the results, the system correctly diagnoses the provided ECG signal is 90% sensitivity (SN) and specificity (SPE) [21]. R. Smíšek, J.Hejč, et al(2017) suggested a technique for classifying ECGs into four categories noise sounds [P], AF [A], other rhythmic [O], and regular pulse [N]. SVM technique was used in the model at two separate phases. In the first stage, SVM was used to extract global features from that ECG dataset. In the second stage, the characteristics from the previous phase were used to train this second SVM classifier. Both classifiers were evaluated using the cross-validation method. The cumulative F1 score of the approach in Phase II of the challenge was 0.81 and 0.84, respectively, in the hidden challenge dataset and training set [22]. W. F. Wang et al. (2017) devised a method for detecting binge drinking. Signals from (ECG), intoxilyzers, and a photoplethysmograph were collected using three sensors (PPG). Participants' alcohol intake levels before and after drinking were measured using intoxilyzers. All signals are pre-processed, segmented, feature extracted to machine learning to provide ECG and PPG training and test datasets. The constructed model for the identification strategy utilizing the SVM algorithm was quick and accurate when using ECG, PPG, and alcohol intake data. The trained SVMs' recognition performance was validated by employing a test dataset and also the training dataset for training. The suggested classifiers have an average accuracy of 95% in identifying objects. Various feature combinations were evaluated to get the best technical arrangement. A technical setup based on PPG is chosen for constructing intelligent and wearable devices that detect driving and intoxication since PPG with ECG characteristics produces equivalent classification findings and PPG features are easy to get [23]. C. Venkatesan, P.Karthigaikumar et al. (2018), To categorize participants into normal and anomalous, ECG flag which was before and SVM-based arrhythmic beat categorization is used. A postponed mistake normalization learning management system LMS adaptive filters is used in

ECG flag pre-processing to achieve high speed and decrease inactivity plan with fewer computing components. White commotion expulsion is essentially emphasized since the flag preparation technique is designed for far-off healthcare frameworks. For HRV highlight extraction, DWT is coupled to the pre-processed flag, and arrhythmic beat categorization is performed using machine learning algorithms. The Classification method and other prominent classifiers were used to classify beats on a noisy evacuated highlighting extricated flag in this work. According to the results, the SVM classification outperforms other machine learning-based classifiers [24]. J.Liu, S. Song, G et al. (2019) developed a CNN-based ECG arrhythmia classification method. The tested CNN methods to SVM with linear classifier analyses (LDA) methods (SVM). All cardiac arrhythmia beats were taken from the MIT-BIH Heart Rhythm Dataset, which was divided into five categories based on the criteria set out by the Association for the Advancement for Clinical Instrumentation. The training and testing sets are from separate persons, and the classification accuracy greater than is 90% [25]. Table III shows the comparison of the mentioned papers that used SVM. The following set of equations describes how to obtain the separation hyper-plane.

TABLE III : SVM-BASED CLASSIFICATION OF HEARTBEAT ALGORITHMS

Author Name	DataBase	Purpose	Models	Accuracy
F. A. Elhaj, N. Salim et al. (2016)	MIT-BIH arrhythmia dataset	ECG signal categorization with great precision	PCA-DWT-ICA-HOSNN-SVM-RBF	Accuracy is SVM (RBF)=98.9% (NN)=98.9%
V. R. Arjunan (2016)	MIT-BIH arrhythmia dataset	Categorize ECG by an automated system	SVM	Accuracy=90%
R. Smešek, J. Hejč, et al (2017) et al.	Hidden dataset of 2017 PhysioNet/ CinC Challenge	An advanced method for automatic classification ECG	SVM-RBF	F1-measure=0.81
W. F. Wang et al. (2017)	Collect data by sensors	Recognize drunk driving by ECG and PPG	SVM	Accuracy=95%
C. Venkatesan, P. Karthigai kumar et al. (2018)	MIT-BIH arrhythmia dataset	Classifier with low computational complexity	SVM	Accuracy=96%
J. Liu, S. Song, G et al. (2019)	MIT-BIH arrhythmia dataset	To perform real-time analysis, a robust and efficient model is required ECG	SVM+CNN+LDA	Accuracy >90%

$$f(x) = (w \cdot \phi(x)) + b \quad (4)$$

$$RsVM(c) = c \frac{1}{N} \sum_{i=1}^N L_{\epsilon}(y_i y_i^{\Delta}) + \frac{1}{2} W^T \cdot W \quad (5)$$

$$L_{\epsilon}(y_i y_i^{\Delta}) = |y_i - y_i^{\Delta}| - \epsilon |y_i - y_i^{\Delta}| \geq \epsilon \quad (6)$$

$$y^{\Delta} = f(x) = \sum_{i=1}^N (a_i - a_i^*) K(x_i, x) + b \quad (7)$$

$$K(x_i, x) = \exp\left(\frac{-1}{\delta^2} (x_i - x_j)^2\right) \quad (8)$$

C. Classification Using Long Short Term Memory (LSTM)

Any signal's temporal characteristic is a time-series feature that evolves. The Recurrent Neural Network (RNN) and the LSTM approach are two deep learning algorithms that can cope with time-series information. Both have been used in a variety of settings, including ECG categorization. Figure 4 shows the Long short-term memory. Ö.Yildirim(2018) has proposed a modern show named DBLSTM-WS (Profound bidirectional LSTM network-based wavelet groupings) to classify pulse. The MIT-BIH arrhythmia database was utilized to induce five sorts of heartbeats. These five sorts are Typical Sinus Cadence (NSR), Paced Beat (PB), Cleared out Bundle Department Piece (LBBB), Ventricular Untimely Withdrawal (VPC), and Right Bundle Department Piece (RBBB). The Accuracy is 99.39% [26]. H.Assodiky, I.Syarif et al. (2018) have proposed a framework to classify Arrhythmia built on the LSTM. AdaDelta was utilized as a versatile learning rate strategy to extend the introduction of the arrange. This approach compares to the LSTM without AdaDelta, which had a tall precision of 98% for preparing data and 97% for test information [27]. E. K. Wang et al. (2019) have proposed a cost-sensitive classifier built on stacked de-noising Auto Encoders Soft Max relapse layer SDAE and Bi-LSTM (bidirectional LSTM) to classify ECG signals. This demonstration was connected to the supraventricular arrhythmia database (SVDB), Noise Stress Test database (NSTDB), and MIT- BIH Arrhythmias Database, individually. The Accuracy result of this approach was 98.25% [28]. J. Gao, H. Zhang et al. (2019) have proposed LSTM with Central Misfortune (FL) to demonstrate preparing to create the preparing impact by affecting numerous ordinary ECG information. This approach has gotten the information from the MIT-BIH arrhythmia database. The precision of this approach was 99.26%. This result clarifies that the model may be a great arrangement to the imbalanced datasets [29]. A. Sharma et al. (2020) have proposed a classifier that combined Fourier-Bessel (FB) with LSTM. This classifier utilized the MIT-BIH database to identify standard and unpredictable ECG. The exactness of this classifier is 90.07% [30]. Table IV shows the comparison of the mentioned papers that used LSTM. Show Fig 6 Long short-term memory.

TABLE IV: LSTM CLASSIFICATION OF HEARTBEAT ALGORITHMS

Author Name	DataBase	Purpose	Models	Accuracy
Ö.Yildirim 2018	MIT-BIH arrhythmia dataset	Classify five types of ECG	DBLSTM-WS	Accuracy= 99.39%.
H.Assodik y,I.Syarif et al. 2018	MIT-BIH arrhythmia dataset	Classify Arrhythmia	LSTM+ AdaDelta	Accuracy= 97%
E. K. Wang et al 2019	MIT-BIH arrhythmia dataset.	Classify ECG signals	SDAE + Bi-LSTM	Accuracy= 98.25%
J.Gao, H. Zhang, et al.2019	MIT-BIH arrhythmia dataset.	Classify Arrhythmia	LSTM+FL	Accuracy= 99.26%.
A. Sharma et al. 2020	MIT-BIH arrhythmia dataset	Classify ECG	FB+LSTM	Accuracy= Over 90%

The equations for the cell state, candidate cell state, and the final output LSTM:

$$c_t^* = \tanh(w_c [h_{t-1}, x_t] + b_c) \quad (9)$$

$$c_t = f_t * c_{t-1} + i_t * c_t^* \quad (10)$$

$$h_t = O_t * \tanh(c_t) \quad (11)$$

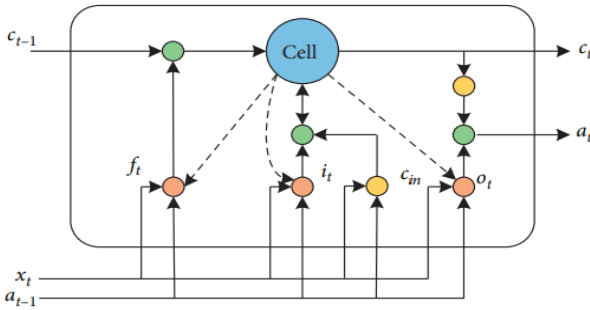


Fig 6: Long short-term memory [29].

IV. CONCLUSION

The classification of ECG signals is crucial in identifying normal and irregular heartbeats. Increasing the ECG classification's ACC is a difficult challenge to solve. It has piqued curiosity for more than a decade, prompting the development of several techniques. The most current techniques are evaluated in terms of the dataset, method, Purpose, and Result in this study. Table II out lines ECG signal analysis variance techniques. The authors recommend employing a CNN-based hybrid model. Table III used Support Vector Machine SVM is a learning system with several benefits. It has to deal with pattern classification and data analysis. It uses a linear discriminate function for classification. Table IV Long short term memory (LSTM) component since the LSTM is better adapted to time - series, it can recognize patterns in temporal characteristics. After that, the system can predict unknown ECG signals.

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